L-2/T-II/CSE **Date: 16/05/2026**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-II B. Sc. Engineering Examinations 2023-2024

Sub: **CSE 219** (Signals and Linear Systems)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. You must answer the **Question No. 1, which contains 45 marks**. Answer any **TWO** questions out of the questions **between 2 and 4**, each of which **contains 30 marks**.

1. (a) Let **(15)**

$$x(t)=\begin{cases}1-|t|, & |t|\le 1,\\ 0, & \text{otherwise.}\end{cases}$$

Sketch the signal

$$y(t)=x(-2(t-1))+x\left(\frac{t+2}{2}\right)$$

You must clearly show the effect of each signal operation on the two components signals before obtaining the final sketch of $y(t)$. In your sketch, you need to properly label both x-axis and y-axis.

(b) Consider the combination of discrete-time LTI systems shown in Figure 1. The input signal $x[n]$ is applied to the overall system, which provides the output signal $y[n]$. The impulse responses $h_1[n]$, $h_2[n]$ and $h_3[n]$ are define as follows **(15)**

$$h_1[n]=\delta[n]-\delta[n-2],\quad h_2[n]=u[n-1],\quad h_3[n]=\delta[n]-\delta[n-1]$$

Suppose the input is x[n] = u[n]. Analyze Figure 1 and answer the following questions.

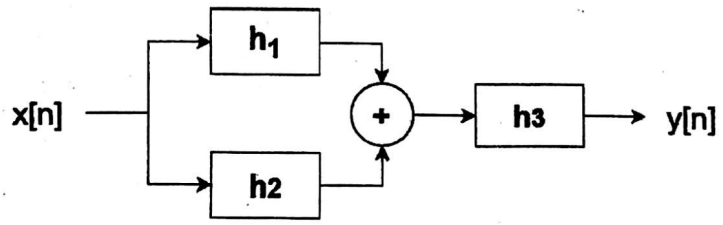


Figure 1: The pictorial representation of the combination of LTI systems for question 1(b).

(i) Express the equivalent impulse response $h_{eq}[n]$ of the overall system.

(ii) Infer the output $y[n]$ for the given input $x[n]$.

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Contd ... Q. No. 1

(c) Determine the Fourier transform of the rectangular function $x_1(t) = rect(t)$. Then apply the properties of continuous Fourier transform and find he Fourier transform of $x_2(t)$ where **(15)**

$$x_2(t) = rect\left(\frac{t-\tau}{T}\right)\cos 2\pi f_0 t$$

2. Suppose, you have two signals $x_1(t) = e^{-at}\,u(t)$ and $x_2(t) = e^{-a|t|}$. Answer the following questions.

(a) Determine the Fourier transform of $x_1(t)$. Then using your answer, determine the Fourier transform of $x_2(t)$. Mention the properties you used in your answer. **(10)**

(b) Evaluate the integral $I(a)$ given below. You have to use Parseval's theorem, your answer from question 2(a) and properties of continuous Fourier transform. **(20)**

$$I(a) = \int_{-\infty}^{\infty} \frac{f^2}{\left(a^2 + (2\pi f)^2\right)^2}\,df$$

3. (a) Suppose **(15)**

$$x[n] = (3-4j)\,e^{j\frac{13\pi}{7}n}$$

is a complex exponential discrete-time signal.

(i) Show whether the real and imaginary parts of $x[n]$ are periodic or not. If periodic, determine their fundamental period.

(ii) Justify carefully whether the phase shift in each part can be interpreted as an integer time shift of a cosine sequence.

(b) The following are the impulse responses of continuous-time LTI systems. Determine whether each system is causal and/or stable. Justify your answers. **(15)**

(i) $h(t) = e^{-4t}u(3-t)$

(ii) $h(t) = e^{-6|t|}$

(iii) $h(t) = te^{-t}u(t)$

4 Let $x(t)$ be the odd square wave

$$x(t) = \begin{cases} A, & 0 < t < \dfrac{T_0}{2}, \\ -A, & -\dfrac{T_0}{2} < t < 0, \end{cases} \qquad x(t + T_0) = x(t).$$

Now define

$$y(t) = x\left(t - \frac{T_0}{6}\right).$$

(a) Show that $y(t)$ still has half-wave symmetry. **(5)**

(b) Find the complex Fourier series coefficients Y_k from those of $x(t)$. **(10)**

(c) Draw the magnitude and phase spectra of Y_k (upto $k = 5$ is sufficient). **(10)**

(d) Compare the trigonometric Fourier series of $x(t)$ and $y(t)$. **(5)**

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SECTION – B

There are **FOUR** questions in this section. **Question No. 5 is mandatory.**
Answer any **TWO** questions from the **Question No. 6, 7, and 8.**

5. (a) A discrete-time signal is given as: **(15)**

$x[n] = \{1, 2, 3, 4, 3, 2\}$

Calculate the 8-point Discrete Fourier Transform (DFT) of the sequence using the Radix-2 Decimation-in-Time (DIT) FFT algorithm, showing all intermediate stages with proper butterfly diagram.

(b) Two discrete-time sequences are given as: **(10)**

$x[n] = \{1, 2\}$ and $h[n] = \{1, 1\}$

Demonstrate the step-by-step process of computing the linear convolution of $x[n]$ and $h[n]$ using the naive Discrete Fourier Transform (DFT) method, performing all calculations to obtain the final time-domain result.

(c) A sound engineer is restoring a vintage vinyl recording. He has digitised the record at a sufficient sampling rate (no aliasing). He now needs to convert the stored digital samples back to a smooth analogue audio signal for playback. He has three reconstruction methods available: an ideal sinc-based low-pass filter, a Zero-Order Hold (ZOH) circuit, and a linear interpolation module. His hardware cannot implement non-causal filters, and his system requires a method that can be directly embedded into a real-time Digital-to-Analog Converter (DAC) with no delay buffer. Based on the scenario above, answer the following questions. **(5+5)**

(i) Compare the impulse responses of the three reconstruction methods mentioned above.

(ii) Justify which single reconstruction method the engineer should choose for his real-time DAC.

6. (a) Solve the following differential equation using the Laplace Transform: **(12)**

$$\frac{d^2y(t)}{dt^2} + 7\frac{dy(t)}{dt} + 12y(t) = \frac{dx(t)}{dt} + 2x(t)$$

with input $x(t) = u(t)$ and initial conditions $y(0) = 0$, $y'(0) = 0$.

(b) Find the Laplace Transform $X(s)$ and its Region of Convergence (ROC) for the following signal $x(t)$. Also, sketch the ROC in the s-plane, clearly marking all poles. **(13)**

$$x(t) = |t|e^{-2|t|} - e^{-t}\cos(3t)u(t) + t^2e^{-4t}u(t)$$

(c) Is it possible for a system to be: **(5+5)**

(i) Causal but unstable?

(ii) Stable but non-causal?

For each case, explain with reasoning based on the Region of Convergence (ROC) and support your answer with an example.

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7. (a) Using the Radix-2 Cooley-Tukey 2D Matrix Formulation, compute the Discrete Fourier Transform of the sequence x[n] = {1, 2, 3, 4, 9, 0, 7, 8}. **(15)**

Follow the steps below for step-by-step calculation:

(i) Arrange the sequence into the standard 2 × 4 column-major matrix, separating even and odd indices. Compute the 4-point DFT along each row. Show the exact complex values of the resulting intermediate matrix G.

(ii) Determine the exact rectangular form $(a + jb)$ of the Cooley-Tukey twiddle factors $W_8^{n_1 k_2}$, where $n_1 \in \{0, 1\}$ is the row index and $k_2 \in \{0, 1, 2, 3\}$ is the column index. Multiply the intermediate matrix G element-wise by these twiddle factors to produce matrix H.

(iii) Compute the 2-point DFT (butterfly operations) down each column of matrix H. Read the matrix in row-major order to state the final sequence $X[k]$. Group your final answers neatly into real and imaginary parts.

(b) Derive the formulation of an N-point Discrete Fourier Transform (DFT) as a linear convolution using the Chirp-Z identity $kn = \frac{n^2}{2} + \frac{k^2}{2} - \frac{(k-n)^2}{2}$ to maintain $O(N \log N)$ complexity for prime lengths, and outline the practical implementation of this convolution via Fast Convolution, explaining the requirement of padding $M \geq 2N - 1$. **(10)**

(c) State the Nyquist-Shannon Sampling Theorem and analyze the behavior of the sampled spectrum $X_p(j\omega)$ when the sampling frequency ω_s both satisfies and violates the Nyquist condition. Illustrate your answer with appropriate spectral sketches, and hence briefly explain why perfect reconstruction of the original signal is not possible once aliasing occurs. **(10)**

8. (a) Determine all possible two-sided sequences $x[n]$ that correspond to the following Z-Transform. **(12)**

$$X(z) = \frac{1 + 2z^{-1} - z^{-2}}{1 - \frac{11}{6}z^{-1} + z^{-2} - \frac{1}{6}z^{-3}}$$

For each sequence, clearly state:

(i) The associated Region of Convergence (ROC)

(ii) The resulting mathematical expression for $x[n]$.

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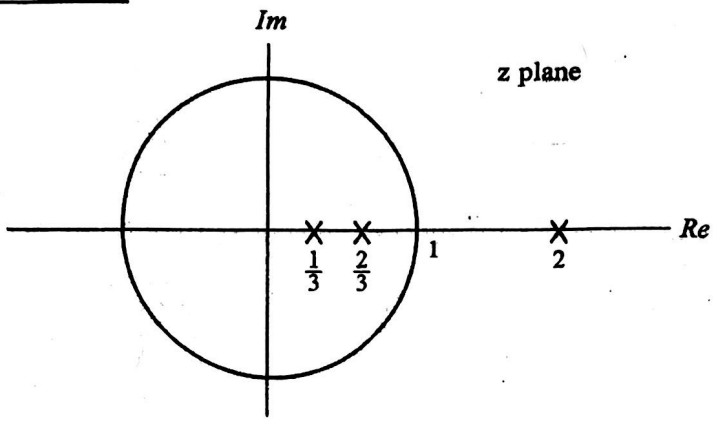


Figure 2: Pole-zero plot of $Y(z)$ for question 8(b).

(b) Figure 2 shows the pole-zero plot of the Z-transform $Y(z)$ corresponding to a discrete-time sequence $y[n]$. Determine what can be inferred about the associated region of convergence (ROC) from each of the following independent statements: **(8)**

(i) $y[n]$ is right-sided.

(ii) The Fourier Transform of $y[n]$ converges.

(iii) The Fourier Transform of $y[n]$ does not converge.

(iv) $y[n]$ is left-sided.

(c) Determine the Z-Transform for the following sequence and sketch the pole-zero plot. **(10)**

$$x[n] = |n| \left(\frac{1}{2}\right)^{|n|}$$

(d) Differentiate between the Laplace Transform and the Z-Transform, and briefly justify why the Z-Transform is preferred for discrete-time systems. **(5)**

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